

MacSync-WCO2X Wi-Fi CO₂ Indoor Air Quality Sensor Datasheet

Product Overview

MacSync-WCO2X is a high-accuracy **Wi-Fi-based CO₂ indoor air quality (IAQ) sensor** designed for continuous monitoring of carbon dioxide levels in smart buildings, workplaces, schools, hospitals, and industrial indoor environments where reliable IP connectivity is available.

Powered by an industrial-grade **NDIR CO₂ sensing module** with automatic baseline calibration, the device delivers precise CO₂ measurements in ppm along with **temperature and humidity monitoring**, providing complete indoor air quality intelligence for ventilation assessment and HVAC optimization.

MacSync-WCO2X communicates directly over **standard Wi-Fi networks (2.4 GHz / 5 GHz)** to **on-premise servers, building management systems (BMS), SCADA platforms, or cloud dashboards** using industry-standard protocols such as **MQTT, HTTP, and HTTPS**—without requiring gateways or long-range wireless infrastructure.

Designed for both new installations and retrofit deployments, the sensor supports **battery-operated and externally powered variants**, enabling flexible installation across diverse indoor locations. **BLE-based provisioning via the Macnman Maya App** allows fast configuration, diagnostics, and parameter updates without specialized tools.

From **classroom ventilation compliance and office occupancy insights** to **healthcare safety monitoring and industrial workspace air-quality management**, **MacSync-WCO2X transforms indoor CO₂ data into actionable intelligence**, enabling healthier, more energy-efficient, and compliant indoor environments.





Key Features

High-Accuracy NDIR CO₂ Sensing

Industrial-grade NDIR sensor delivers precise CO₂ measurements in ppm with automatic baseline calibration for long-term stability.

Wi-Fi Based Connectivity

Direct IP communication over standard Wi-Fi networks (2.4 GHz / 5 GHz) without the need for gateways.

Real-Time CO₂ Alerts & Threshold Events

Configurable alerts for ventilation control, indoor air-quality compliance, and occupant safety.

Standard IoT Protocol Support

Supports MQTT, HTTP, and HTTPS for seamless integration with BMS, SCADA systems, and cloud platforms.

BLE Configuration via Maya App

Fast provisioning, diagnostics, and parameter configuration using the Macnman Maya mobile application.

Integrated Temperature & Humidity Monitoring

Simultaneous CO₂, temperature, and humidity measurement for complete indoor air-quality (IAQ) assessment.

Battery-Powered & Externally Powered Variants

Available in battery-operated and 9–36 V DC powered versions for flexible indoor deployment.

Smart Building & HVAC Ready

Enables demand-controlled ventilation and energy optimization based on real-time CO₂ levels.

Secure Wi-Fi Communication

WPA2 / WPA3 authentication with TLS-encrypted data transmission for secure IP-based communication.

Indoor-Optimized Industrial Design

Compact IP-rated enclosure designed for offices, classrooms, hospitals, and commercial indoor environments.

Suited Applications

Smart Buildings & Offices

Continuous CO₂ monitoring to maintain healthy indoor air quality, improve occupant comfort, and optimize energy usage.

HVAC & Ventilation Optimization

Enables demand-controlled ventilation based on real-time CO₂ levels to reduce energy consumption while maintaining IAQ compliance.

Commercial Buildings & Campuses

CO₂ monitoring across offices, corporate campuses, malls, and shared workspaces for centralized facility management.

Warehouses & Logistics Facilities

Maintains safe air quality levels in enclosed storage areas and high-occupancy logistics environments.

Schools & Educational Institutions

Ensures adequate ventilation in classrooms, laboratories, libraries, and auditoriums to support student health and learning outcomes.

Hospitals & Healthcare Facilities

Supports indoor air-quality compliance in wards, ICUs, waiting areas, pharmacies, and diagnostic rooms.

Industrial Indoor Workspaces

Protects worker health by monitoring CO₂ accumulation in factories, production floors, control rooms, and workshops.

Hospitality & Public Indoor Spaces

Improves guest comfort and regulatory compliance in hotels, restaurants, gyms, conference halls, and indoor public areas.

Technical Specifications

General Specifications

Part Order Number	MacSync-WCO2X
Product Brand	Macnman Technologies Pvt Ltd

CO₂ Measurement

CO ₂ Measurement Range	400 ppm – 2,000 ppm 400 ppm – 5,000 ppm
Accuracy	± (50 ppm + 5% of reading)
Repeatability	± 10 ppm
Response Time (T90)	≤ 60 seconds
Accuracy Drift (per year)	± (5 ppm + 0.5% of reading)
Sensor Type	NDIR (Non-Dispersive Infrared) with Auto-Calibration
Calibration	Factory calibrated

Temperature Measurement

Range	-10 °C to +60 °C
Accuracy	± 0.8 °C
Resolution	0.1 °C
Accuracy Drift	< 0.03 °C / year
Sensor Type	Integrated Temperature Sensor

Humidity Measurement

Range	0% to 100% RH (non-condensing)
Accuracy	±3% RH (10%–90% RH), ±5% RH (<10% or >90% RH)
Resolution	0.5% RH
Calibration	Factory calibrated

Wireless Communication

Wi-Fi Standards	IEEE 802.11 b/g/n/ac
Frequency Bands	2.4 GHz & 5 GHz (Dual-Band)
Operating Modes	Station Mode (Client) Access Point Mode (Configuration)
Network Protocols	MQTT, HTTP, HTTPS
Security	WPA2-PSK / WPA3 TLS / SSL Encrypted Communication
IP Configuration	DHCP / Static IP
Typical Indoor Range	Up to 100 m (environment dependent)

LED Status

LED Type	1 Internal Multi-Color LED
Solid Blue	Device is connected with Maya App
Flashing Blue	OTA (Over-the-Air) update in progress
Slow Blinking Green	Attempting to connect to the network

Single Red Blink	A confirmed parameter of MacSync is being modified
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Device Configuration

Configuration	Via Macnman Maya Android App
Uplink Modes	Time Based / Threshold Based / Sampling Based
Stored Records*	30,000 Records
Remote Configuration	JSON Format Data - from server

Supported Cloud & Platforms

Deployment Models	On-Premise Servers Private Cloud Deployments Public Cloud Platforms
IoT Cloud Platforms	AWS IoT Core Microsoft Azure IoT Hub Google Cloud (via MQTT / HTTPS) Custom IoT Platforms
Integration Methods	MQTT Brokers REST / HTTPS APIs Webhooks
Enterprise Systems	SCADA Systems BMS / IBMS Platforms Data Lakes & Analytics Engines

Physical Characteristics

Power Supply	9-36 V DC
Operating Temperature	-40°C to +80°C
Humidity Range	0% to 100% RH (non-condensing)
Ingress Protection	IP65 standard (IP67 optional variant)
Housing Material	Glass filled Nylon
Color	Matte White
Weight	~110g
Mounting Options	Pole Mounting / Wall Mounting

Electrical Characteristics

Input Voltage Range	9-36 V DC (External Supply)
Power Consumption	~100 µA in sleep mode ~150 mA during transmission
Input Impedance	>10 MΩ for voltage inputs

Communication Options

BLE Connectivity	For local configuration and diagnostics via Maya App
Cloud Protocols	MQTT / HTTPS
Serial Interface	RS485 Modbus RTU support (optional)

Certifications & Compliance

WPC	Compliant
RoHS	Conforms to RoHS directive

Security & Encryption

Data Encryption	AES-256 end-to-end encryption
Authentication	OTAA activation modes
Firmware Protection	Secure boot and signed OTA updates

MacSync-WCO2X Variants

VARIANT NAME	CO ₂ RANGE	POWER SOURCE	BATTERY CAPACITY	IDEAL FOR
MacSync-WCO2X1-PO	400-2,000 ppm	External Power (9-36 V DC)		HVAC automation in commercial buildings with stable power
MacSync-WCO2X1-BO	400-2,000 ppm	Battery Powered	19 Ah	Schools, offices & hospitals focused on IAQ compliance
MacSync-WCO2X2-PO	400-5,000 ppm	External Power (9-36 V DC)		Industrial & high-density areas requiring extended CO ₂ sensing

VARIANT NAME	CO ₂ RANGE	POWER SOURCE	BATTERY CAPACITY	IDEAL FOR
MacSync-WCO2X2-BO	400–5,000 ppm	Battery Powered	19 Ah	Retrofit deployments, warehouses & logistics hubs

What's in the Box?

Make sure box contains following items before installations



MacSync



Mounting
Clip



User manual &
Warranty Card



Junction
Box*



Mounting
Screws

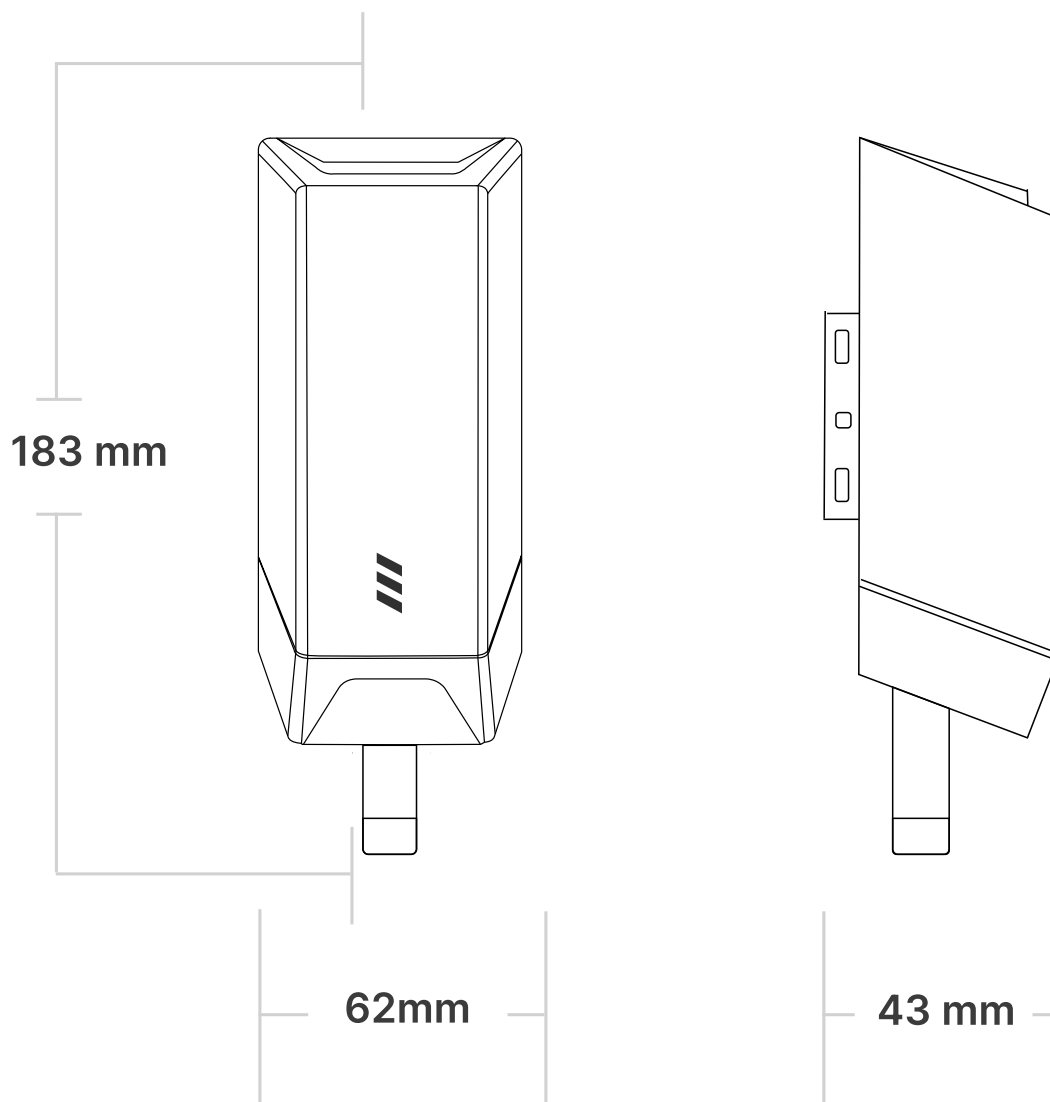


Power
Adapters*

Note

Accessories must be purchased separately & may not be included in the box.

MacSync-WCO2X Dimensions



Frequently Asked Questions

What is the MacSync-WCO2X used for?

MacSync-WCO2X is a high-accuracy Wi-Fi CO2 indoor air quality sensor for continuous carbon dioxide, temperature and humidity monitoring in smart buildings. Typical applications include Smart Buildings & Offices, Schools & Educational Institutions, HVAC & Ventilation Optimization, Hospitals & Healthcare Facilities, Commercial Buildings & Campuses and Industrial Indoor Workspaces.

What does the MacSync-WCO2X measure, and how accurately?

Measurement specifications of the MacSync-WCO2X:

- CO₂: range 400 ppm – 2,000 ppm, 400 ppm – 5,000 ppm, accuracy $\pm$ (50 ppm + 5% of reading).
- Temperature: range -10 °C to +60 °C, accuracy $\pm$ 0.8 °C, resolution 0.1 °C.
- Humidity: range 0% to 100% RH (non-condensing), accuracy $\pm$ 3% RH (10%–90% RH), $\pm$ 5% RH (<10% or >90% RH), resolution 0.5% RH.

How does the MacSync-WCO2X connect and send data over Wi-Fi?

The MacSync-WCO2X connects over IEEE 802.11 b/g/n/ac Wi-Fi on 2.4 GHz & 5 GHz (Dual-Band).

Security: WPA2-PSK / WPA3, TLS / SSL Encrypted Communication. IP configuration: DHCP / Static IP. It sends data using MQTT, HTTP, HTTPS. Typical indoor range is up to 100 m (environment dependent).

How is the MacSync-WCO2X powered?

The MacSync-WCO2X is powered from 9-36 V DC (External Supply).

What is the IP rating and operating temperature of the MacSync-WCO2X?

The MacSync-WCO2X has an ingress protection rating of IP65 standard (IP67 optional variant) and an operating temperature range of -40°C to +80°C. Operating humidity: 0% to 100% RH (non-condensing). Mounting options: Pole Mounting / Wall Mounting.

How is the MacSync-WCO2X configured?

The MacSync-WCO2X is configured with the Macnman Maya Android app. Supported uplink modes: Time Based / Threshold Based / Sampling Based. It can also be configured remotely with JSON data sent from the server.

Which cloud platforms does the MacSync-WCO2X work with?

The MacSync-WCO2X can send data to platforms including AWS IoT Core, Microsoft Azure IoT Hub, Google Cloud (via MQTT / HTTPS), Custom IoT Platforms. Deployment models: On-Premise Servers, Private Cloud Deployments, Public Cloud Platforms. Integration methods: MQTT Brokers, REST / HTTPS APIs, Webhooks.

Related Products and Resources

- Related datasheets: [MacSync LCO2 X1 LoRaWAN CO2 sensor](#), [MacSync-WGSX2 Wi-Fi H2S and NH3 gas sensor](#), [MacSync-W-Odor-X1 Wi-Fi qualitative odor sensor](#)
- Help: [Warranty terms and conditions](#) and [contact Macnman support](#)